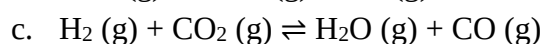
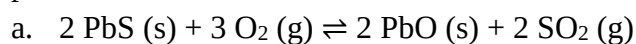


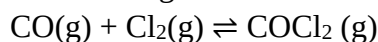
 UTM UNIVERSITI TEKNOLOGI MALAYSIA	Sekolah Pendidikan Profesional dan Pendidikan Berterusan (UTMSpace)	Tutorial 3
		CHEMISTRY II
		FSPC 0034

1. Consider the following equilibrium systems. Predict the direction of the net reaction in each case as a result of decreasing the pressure (increasing the volume) on the system at constant temperature.



*Shift to side with more
num of mol of gas particles*

2. Carbonyl chloride (COCl_2) also called phosgene, was used in World War I was a poisonous gas. The equilibrium concentrations for the reaction between carbon monoxide (CO) and molecular chlorine (Cl_2) to form COCl_2 is given below :

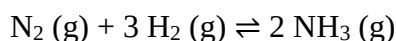


At equilibrium with 77°C , there are $[\text{CO}] = 2.1 \times 10^{-3} \text{ M}$, $[\text{Cl}_2] = 0.065 \text{ M}$, and $[\text{COCl}_2] = 0.42 \text{ M}$. Calculate the equilibrium constant, K_c and then K_p .

3. The equilibrium constant, K_p for the decomposition of carbonic acid (H_2CO_3) in soft drinks is found to be 1.05 at 250°C . If the equilibrium partial pressure of H_2CO_3 and CO_2 are 0.875 atm and 0.463 atm, respectively, what is the equilibrium partial pressure of H_2O at 250°C ?



4. At the start of a reaction, there a 0.429 mol N_2 , 0.0123 mol H_2 , and 6.24×10^{-3} mol NH_3 in a 5.30 L reaction vessel at 375°C . If the equilibrium constant, K_c for the reaction is 1.2 at this temperature, decide whether the system is at equilibrium. If it's not, predict which way the equilibrium will proceed.



5. $\text{H}_2 \text{ (g)} + \text{I}_2 \text{ (g)} \rightleftharpoons 2\text{HI (g)}$

The equilibrium constant, K_c for the above reaction at 200°C is 20.0. At the point during the reaction, a mixture containing 1.0 mol dm^{-3} of H_2 , 1.3 mol dm^{-3} of I_2 and 1.5 mol dm^{-3} of HI is left standing at 200°C .

- Write the reactions quotient, Q_c for the reaction.
- Calculate the reaction quotient, Q_c for the reaction.
- Determine whether the reaction is in the equilibrium or not. Then, determine the net direction the reaction is progressing.

1) (a) shift to left

(b) shift to right

(c) no shift, the reaction will not shift in either direction regardless of pressure change because there are 2 mol of gas particles on the side of the reactants and 2 moles of gas particles on the side of the products

$$2) K_c = \frac{[\text{COCl}_2]}{[\text{CO}][\text{Cl}_2]} = \frac{0.42}{2.1 \times 10^{-3} \times 0.065} = 3076.92$$

$$K_p = K_c (RT)^{\Delta n} \\ = 3076.92 (0.0821)(77+273)^{-1} \\ = 107.08 \text{ atm}^{-1}$$



$$K_p = \frac{(\text{H}_2\text{O})(\text{CO}_2)}{(\text{H}_2\text{CO}_3)}$$

$$1.05 = \frac{(\text{H}_2\text{O})(0.463)}{(0.875)}$$

$$P_{\text{H}_2\text{O}} = 1.98 \text{ atm}$$

$$4) Q_c = \frac{(n_{\text{NH}_3})_0^2}{(n_{\text{H}_2})_0^3 (n_{\text{N}_2})_0} = \frac{6.24 \times 10^{-3^2}}{0.0123^3 \times 0.429} = 48.77$$

$$K_c = 1.2$$

$Q_c > K_c$, since $Q_c \neq K_c$, the system is not equilibrium, it will shift to left until K_c is achieved



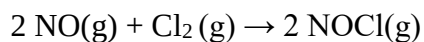
$$5) \text{a) } Q_c = \frac{[\text{HI}]^2}{[\text{H}_2][\text{I}_2]}$$

(c) $Q_c \neq K_c$, the reaction is not equilibrium.

The net direction will be shifted to the right until Q_c is achieved

$$\text{(b) } Q_c = \frac{1.5^2}{(1.0)(1.3)} \\ = 1.731$$

6. Nitric oxide (NO) reacts readily with chlorine gas as follows:



Given pressures of nitric oxide (NO) is 0.16 atm, chlorine (Cl₂) is 0.23 atm and NOCl is 0.56 atm.

- a. Determine the equilibrium constant, K_p , of the reaction.
 - b. At 43 °C temperature, calculate the K_c of the reaction.
 - c. Determine the K_c for the reaction $5 \text{NOCl (g)} \rightarrow 5 \text{NO (g)} + 5/2 \text{Cl}_2 \text{(g)}$
7. The equilibrium constant, K_p for the decomposition of phosphorus pentachloride gas to phosphorus trichloride gas and chlorine gas is 1.00 at 250 °C.
- a. Write down the balanced chemical equation including their physical state and the equilibrium constant in terms of K_p for this reaction.
 - b. Calculate the partial pressures of all reacting species at the equilibrium if the initial pressure partial pressure of phosphorus pentachloride is 0.25 atm flask at 250 °C.
8. Aurumn (III) chloride, (AuCl₃), has a solubility of 54 g/100 mL of water. Determine the solubility-product constant, K_{sp} , of AuCl₃ in water.
9. Iron (III) oxide, (Fe₂O₃), has a solubility of 2.3×10^{-3} g/10 mL of water. Determine the solubility-product constant, K_{sp} , of Fe₂O₃ in water.

$$6) a) K_p = \frac{P(\text{HCl})^2}{P(\text{NO})^2 \times P(\text{Cl}_2)} = \frac{0.56^2}{0.16^2 \times 0.23} = 53.26 \text{ atm}^{-1}$$

$$b) K_p = K_c RT^{\Delta n}$$

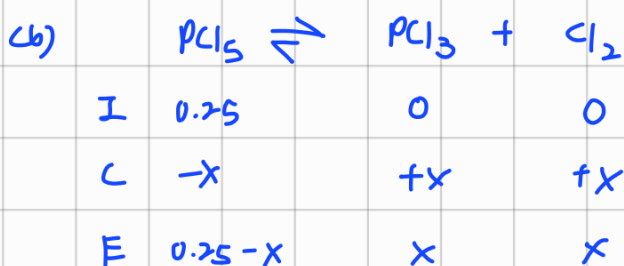
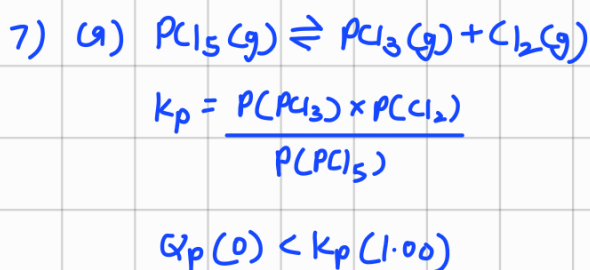
$$53.26 = K_c (0.0821 \times 316)^{-1}$$

$$K_c = 1381.78$$

$$c) K_{c_2} = K_c^{-1(2.5)} \Rightarrow \text{inverse and multiply } \frac{5}{2}$$

$$K_{c_2} = (1381.78)^{-2.5}$$

$$= 1.41 \times 10^{-8}$$



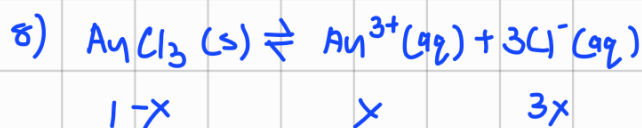
$$1.00 = \frac{x^2}{0.25 - x} \quad P(\text{PCl}_3) = P(\text{Cl}_2) = 0.207 \text{ atm}$$

$$0.25 - x = x^2 \quad P(\text{PCl}_5) = 0.25 - 0.207$$

$$x = 0.207 \quad = 0.043 \text{ atm}$$

$$\text{Solubility} = \text{m. solubility} \times \text{m.h.}$$

$$\frac{\text{g}}{\text{L}} \quad \frac{\text{mol}}{\text{L}} \quad \frac{\text{g}}{\text{mol}}$$



$$\text{m.m. of AuCl}_3: 303.5$$

$$\text{Molar solubility} = \frac{\text{Solubility}}{\text{m.m.}}$$

$$= \frac{540}{303.5}$$

$$= 1.78 \text{ M/L}$$

$$x = 1.78$$

$$K_{sp} = [\text{Au}^{3+}][\text{Cl}^-]^3$$

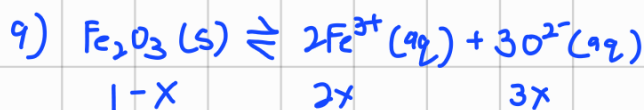
$$= x(3x)^3$$

$$= 27x^4$$

$$= 27(1.78)^4$$

$$= 271.05 \text{ M}^4$$

$$\text{solubility} = \frac{54\text{g}}{100\text{gH}_2\text{O}} \times \frac{1000\text{mL}}{1\text{L}} = 540 \text{ g/L}$$



$$\text{m.m. of Fe}_2\text{O}_3: 160$$

$$\text{molar solubility} = \frac{0.023}{160}$$

$$= 1.44 \times 10^{-4}$$

$$x = 1.44 \times 10^{-4}$$

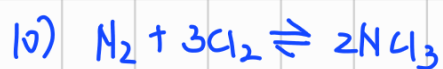
$$K_{sp} = [\text{Fe}^{3+}]^2[\text{O}^{2-}]^3$$

$$= (2x)^2(3x)^3$$

$$= 108x^5$$

$$= 108(1.44 \times 10^{-4})^5$$

$$= 6.69 \times 10^{-18}$$



$$K_c = \frac{[NCl_3]^2}{[N_2][Cl_2]^3} = \frac{0.5^2}{0.15 \times 0.25^3} = 106.67$$



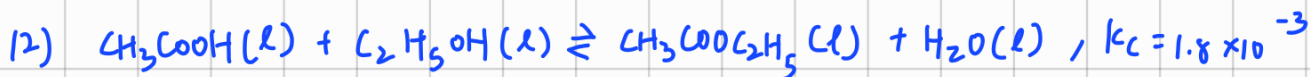
$$K_c = \frac{[NH_3]^2}{[N_2][H_2]^3} = 2.4 \times 10^{-3}$$

ci) Equation $\div 3$

$$K = K_c^{\frac{1}{3}} \\ = (2.4 \times 10^{-3})^{\frac{1}{3}} \\ = 0.134$$

cii) Equation is reversed, $\div 2$

$$K = K_c^{-1 \times \frac{1}{2}} \\ = K_c^{-\frac{1}{2}} \\ = 20.41$$



$$K_c = \frac{[CH_3COOC_2H_5]}{[CH_3COOH][C_2H_5OH]}$$

No K_p because all is liquid, $K_c < 1$, favors reactant



$$K_c = \frac{[PCl_3][Cl_2]}{[PCl_5]} \quad K_p = \frac{(P_{PCl_3})(P_{Cl_2})}{(P_{PCl_5})}, \quad K_c \approx 1, \text{ significant (either products or reactants)}$$



$$K_c = [CO_2][NH_3]^2$$

$$K_p = (P_{CO_2})(P_{NH_3})^2$$

